

$\#$ monotone and even drawing $\Rightarrow$ h has $\#$ monotone embedding (planar)

not system, $\#$ cont. of points

- lemma 2:

Γ of h with bound edge e

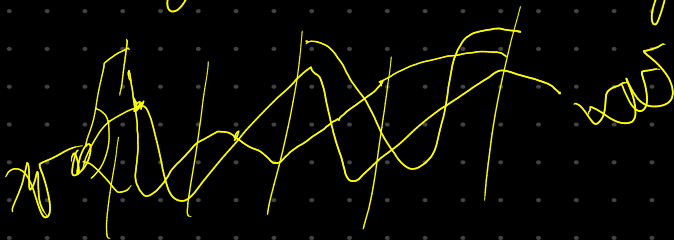
$\Rightarrow e$ can be redrawn (without)

remains Γ changes

$\downarrow$
pos of e
in Γ

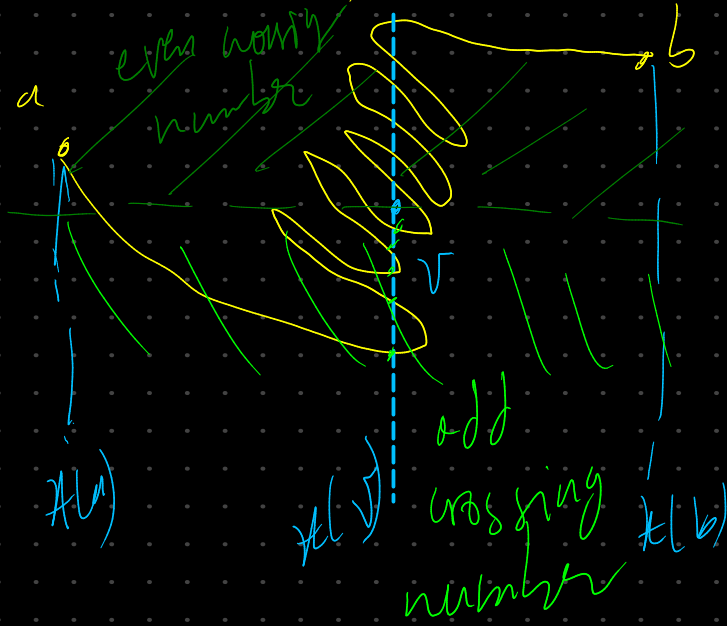
 $e = (v, w)$

$\Rightarrow$ e is $\#$ -monotone
parity of crossing betw e and
any other edge of h has not changed



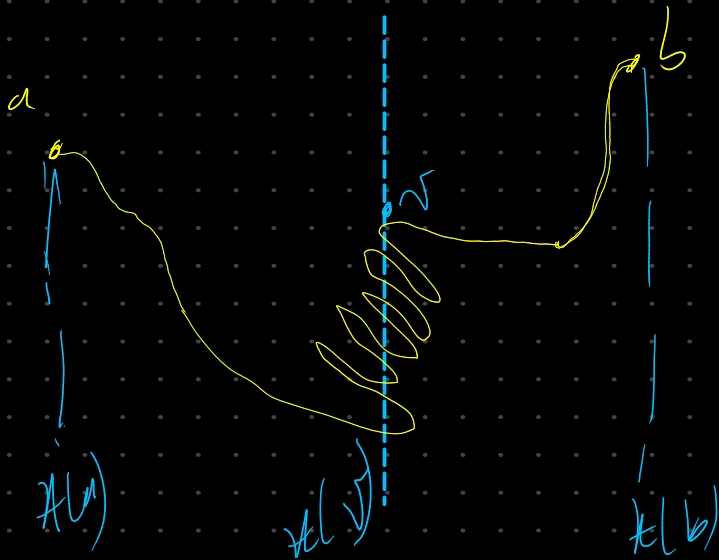
Proof: $e = ab$, let $v \in V(G)$ arbitrary vertex

$$\kappa(a) < \kappa(v) < \kappa(b)$$

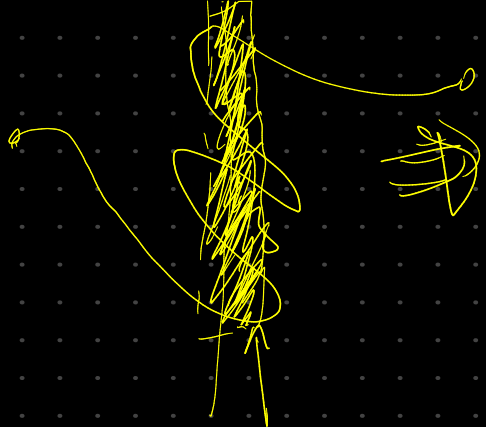


Since e connects a to b it has to cross $\kappa(v)$ odd number of times.

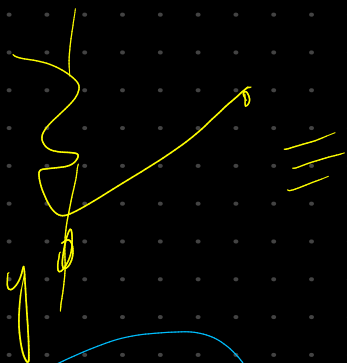
e moves an even number of times across v .



no crossing is left on the axis $\kappa(v)$
Can we propagate along the $[\kappa(a), \kappa(b)]$?



fine that all nodes are below z .



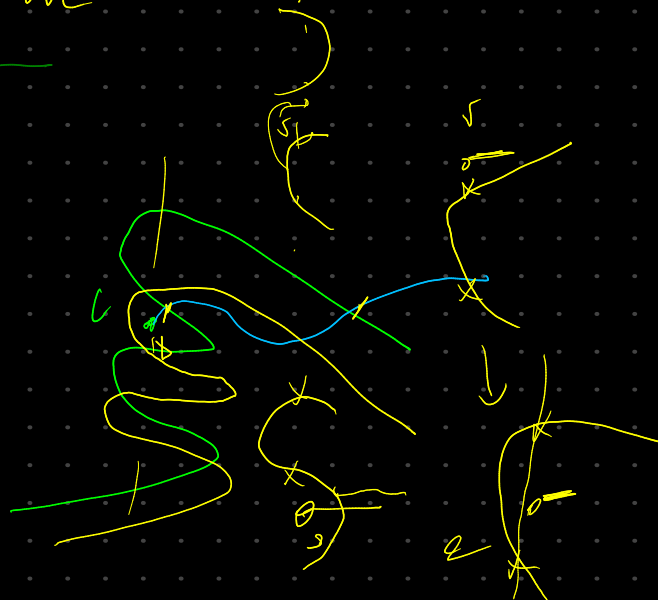
if we have it $xz \in V(w)$
 $x(z) \in [x(w), x(L)]$

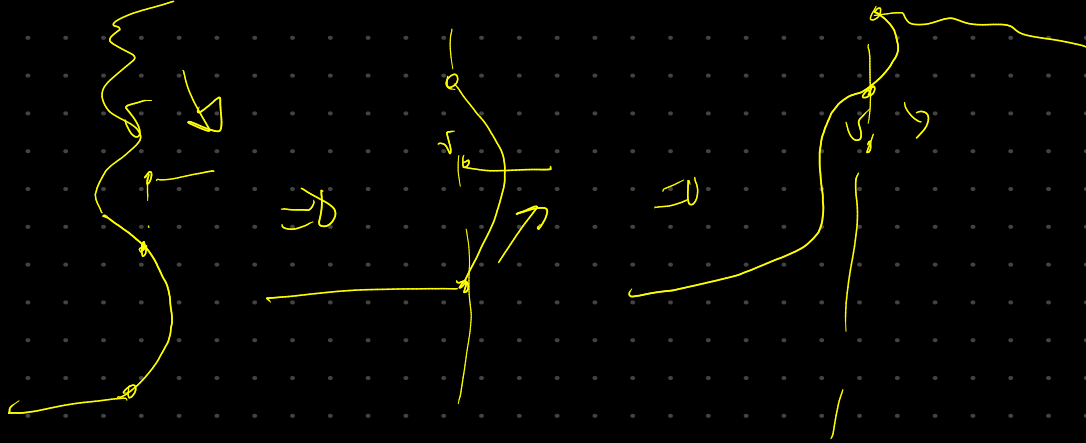
Then $y(z) \geq \max_{e(x(z))} y$

$y(z) \leq \min_{e(x(z))} y$



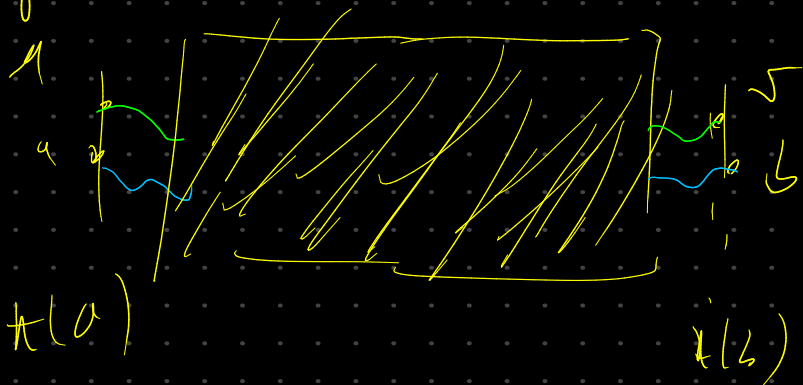
As long as it doesn't cross the vertex above and below it, then we have same parity





continuation of the proof

the issue with parity of crossing is present when one of the endpoints of an edge is strictly in $\int \pi(a), \pi(b) \cap$, since the case where $\pi(a) = \pi(v)$, $\pi(b) = \pi(w)$ changing the shape of e won't be a problem by theorem 1



for the case where $\pi(a) < \pi(v) < \pi(b)$

Moving one crossing to the other side changes the rotation of endpoints if e was below v and below its incident edge then

it will up and it needs an other crossing
 to move to the other side before
 having the rotation and since the
 other side of the edge didn't change
 the relative order in $y(b)$ (suppose we
 add a point representing crossing of the edge
 with $y(b)$) then in shift of even
 crossing $\Rightarrow$ same ^{relative} order in $x(a)$ and
 since still same order in $x(b)$ and
 by theorem 1 same parity

More like
 proof of theorem 1

and we can replace
 by x -coordinate
 v_i v^0

← Strong Hanani-Tutte for
 Monotone Drawings =

TH2) h

x -monotone
drawing

(e.e.)
independent
rows evenly



h has an
 x -monotone emb.
with each vertex
keeping it's x -coord.

Missing

TH1) h

x -monotone
and even
drawing

sub
(relaxed)
constraint

x -monotone emb.
with
 $\forall v \in V(h)$ keeps their
 x -coord.

rotation sys stays
the same

It's a useful relaxation so we don't
really care about keeping the same
rotation system in our problem. let's
see if this relaxation will have easier
ways of building the x -monotone
embedding.

$h = (V, E)$, $h' = (V', E')$
 $h < h'$ if $|V| < |V'|$ or if $|V| = |V'|$ and $|E| < |E'|$

Minimal relation counterexample with respect to this relation
 { why introduce the relation?
 counterexample on what
 what's $h < h'$ to begin with?

HT Then

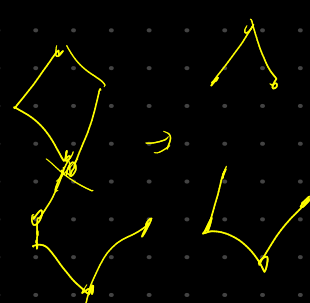
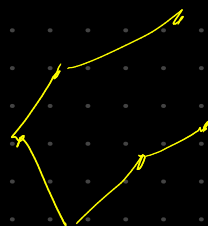
h with independent edges crossing
 even number of times $\Rightarrow$ planar

minimal counterexample : counterexample

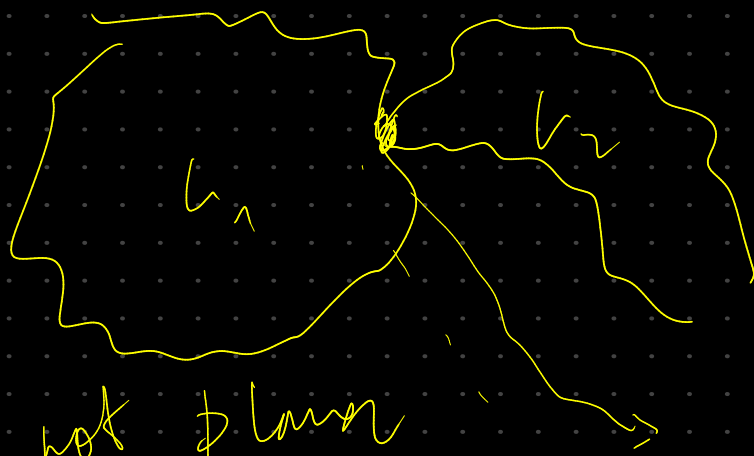
if it's not 2 connected
 (removing a vertex leaves
 h connected)
 then h can be cut

with min $|V(h)|$

$h \in \mathcal{H}$
 not sat'fy the counterexample

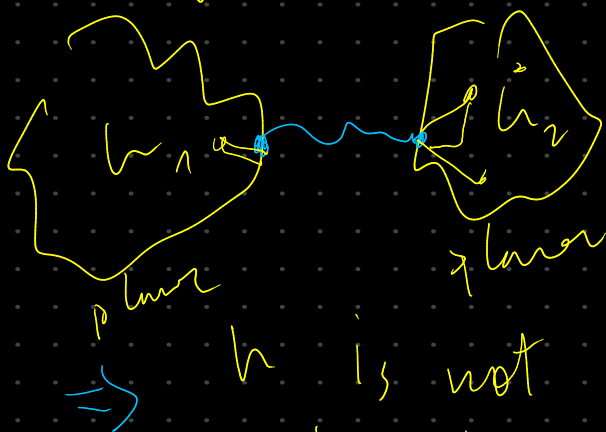


the minimal counterexample for 1H theorem is that we can find h where indep. edges crosses evenly but it's not planar.



$$h = h_1 \cup h_2$$

h is not planar
it can be subdivided to h_1 and h_2 , one of the subgraphs can be not planar.



and we can merge them.

$\Rightarrow h$ is not a counter-example

$\Rightarrow$ min. counterexample must be 2-connected.

The trick is to check if h is not 2-connected
the issue is that even if it's not 2-connected
the merge requires a redrawing that does not maintain monotonicity.

Lemma 3: G is Minimal C.e. of TH 2

connected

G

If G has a cut-vertex a
and $G - \{a\}$ has a component
 H such that $x(a) < x(v)$
 $\forall v \in V(H)$

G has no
 $H \subset G$
 $a, b \in V(G) \setminus V(H)$
s.t. $x(a) < x(v) < x(b)$
 $\forall v \in V(H)$,

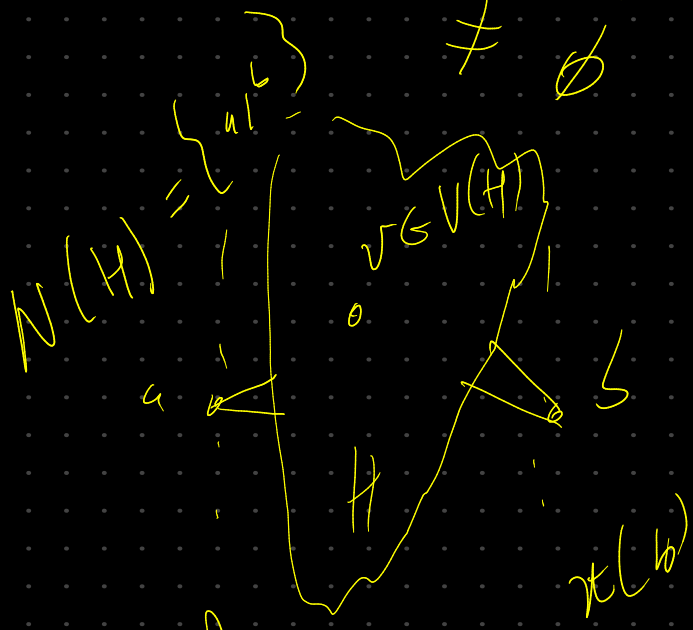
$N(H) = \{a, b\}$,

and $V(G) \setminus (V(H) \cup \{a, b\}) \neq \emptyset$

$\Rightarrow H$ has one vertex
 b , and G has no edge
 ac with $x(b) < x(c)$.

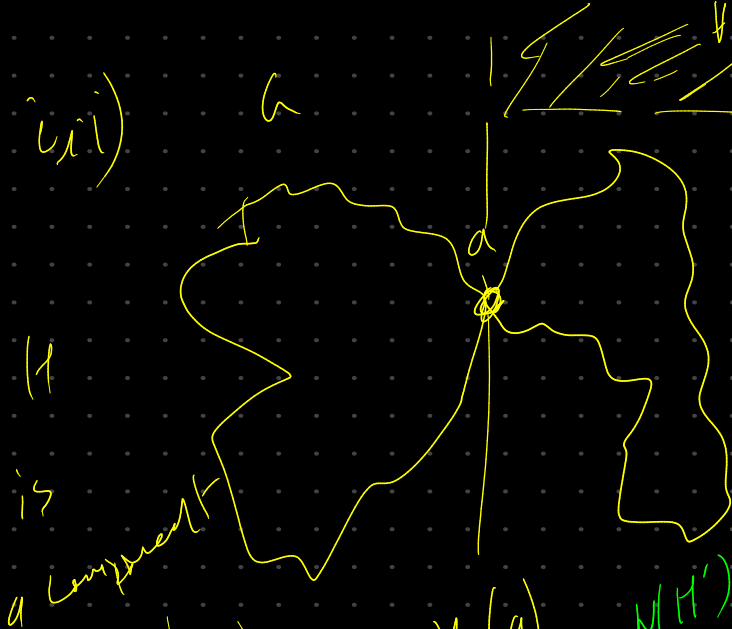
Also, in this case
 G has no induced
 $H' \neq \emptyset$ s.t.
 $x(a) < x(v) < x(b)$
 $\forall v \in V(H')$,
 $a \in N(H') \neq \{a\}$

there are still vertices
that are not
bounded
by a, b
and $x(v) > x(b)$
 $\forall v \in N(H') \setminus \{b\}$



$x(a)$

If I understand
we try to find the boundaries
of G in x -axis
where all $V(G) \setminus \{a, b\}$
contains both $x(a)$, $x(b)$



$v \in V(H)$

$\rightarrow V(H) = \{b\}$

$\& a \in E(H)$ with

$\pi(b) < \pi(a)$

$\& H$ has no induced

$H' \subset H$



with $\pi(a) < \pi(v) < \pi(b)$

$\forall v \in V(H')$

- This is the component that is split if we remove a .

$H' = H \cup \{a\}$

$\pi(v) \in [\pi(a), \pi(b)] \quad \forall v \in V(H')$

with not $N(H')$ composed of just a

$a \in N(H') \neq \{a\}$ and

$\pi(v) > \pi(b) \quad \forall v \in N(H') \setminus \{a\}$

Proof): i) if h is not connected then
 none of its components are counterexamples
 (h is the min) we can stack the
 components vertically and form a
 planar embedding of h . (contradiction)

ii) h is min

$h = V(H) \rightarrow$ have embedding
 $\left\{ h[V(H) \cup \{a, b\}]^+ \right\}$

because both smaller than h

→ start with $h[V(H) \cup \{a, b\}]$, let's
 deform the crossing free drawing to become
 very flat?

If $ab \in E(h) \Rightarrow$ insert the drawing into

the drawing $h = V(H)$ near the edge

ab without adding crossing $\leftarrow h[V(H) \cup \{a, b\}]$
 is planar



$h = H \Rightarrow$ planar

which give a contradiction

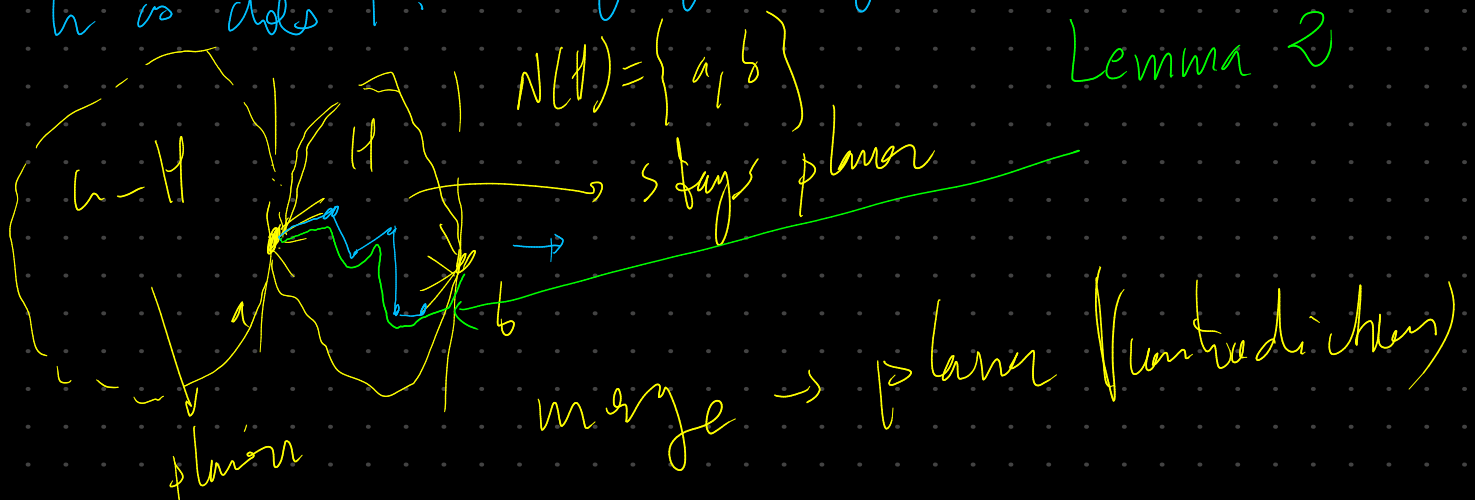
What about $ab \in E(G)$ (ii) doesn't state that there is an edge ab)

They claim that they know how to add ab without introducing ^{int.} crossings.
if that's the case then
 $(G - V(H)) \cup \{ab\}$ has fewer vertices than $G \Rightarrow$ has embedding $\Rightarrow$ same as $ab \in E(G)$

let P be any a, b -path with interior vertices in H . By suppressing the interior vertices we can consider the path a rounded edge.

(This works because $v \in H \Rightarrow x(a) < x(v) < x(b)$)

by lemma 2 we can draw x -monotone Γ edge with the same parity of crossing with all edges of G as does P .



iii) H is a component of $G - \{a\}$

so $\kappa(a) < \kappa(v) \quad \forall v \in V(H)$

if $|V(H)| > 1$, let $b \in V(H)$ s.t. $\kappa(b) > \kappa(v) \quad \forall v \in H$

$H := H - b$ from ii) if $H \subset G$ with

$$\kappa(a) < \kappa(v) < \kappa(b) \quad \forall v \in H, \quad N(H) = \{a, b\}$$

$$\underbrace{V(H)}_{\neq \emptyset} \Rightarrow V(G) \setminus (V(H) \cup \{a, b\}) \neq \emptyset$$

$$V(G) \setminus (V(H - b) \cup \{a, b\}) \neq \emptyset$$

→ TODO:

iii) of Lemma 3

Lemma 4 if possible